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Exercise Sheet 1

Lecture "Cybersecurity"

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Practical Exercise: Classical Cryptography

Terminology

cipher Method/algorithm used for encryption

ciphertext encrypted message

plaintext decrypted message (clear text)

alphabet Set of symbols used for encryption and decryption

substitution cipher A cipher that replaces symbols with other symbols

transposition cipher A cipher that rearranges the order of symbols in the plaintext

Shift Cipher - Caesar

A well-known cipher is the Shift Cipher, also known as Caesar. Caesar belongs to the substitution ciphers. Each letter of the plaintext is shifted by a fixed number in the alphabet. The key 3 means that the letters of the plaintext are shifted by 3 in the ciphertext. So A becomes D, B becomes E, and C becomes F.
Each letter in the alphabet can also be assigned a number, see Table 1.
A Caesar cipher can formally be defined as follows:

> Z_{26} is the set of numbers from 0 to 25; the alphabet is assigned as described in Table 1.
> Let x, y, k $\in Z_{26}$
> Encryption with key k of plaintext letter x:
> $e_k(x) = x + k26$
> Decryption with key k of ciphertext letter y:
> $d_k(x) = y - k26$

Exercise 1.

1	E	17,4%	7	T	6,15%	13	G	3,01%	19	K	1,21%	25	Y	0,04%
2	N	9,78%	8	D	5,08%	14	M	2,53%	20	Z	1,13%	26	X	0,03%
3	I	7,55%	9	H	4,76%	15	O	2,51%	21	P	0,79%	27	Q	0,02%
4	S	7,27%	10	U	4,35%	16	B	1,89%	22	V	0,67%			
5	R	7,00%	11	L	3,44%	17	W	1,89%	23	ß	0,31%			
6	A	6,51%	12	C	3,06%	18	F	1,66%	24	J	0,27%			

Table 1: Deutsche Häufigkeitsverteilung

1. Install Cryptool 2 on your system (`https://www.cryptool.org/de/ct2/downloads`)

2. Encrypt the following plaintext using key $k = 5$ (Shift Cipher) in Cryptool: *Sehr geehrte Damen und Herren, wilkommen bei der Vorlesung fuer Cybersicherheit. Wir starten mit einer klassischen Chiffre Caesear. Und nun ein schoenes Zitat von Gaius Julius Caesar. Das beste Glück, ein schöner Blick, ein kluger Scherz, ein redlich Herz.*

3. Now perform a frequency analysis for the plaintext and ciphertext using Cryptool. The frequency of each symbol (A–Z) is counted and related to the total message length.

4. Compare the resulting bar charts (frequency analysis).

5. Now take various German texts[1] and run a frequency analysis using Cryptool. Compare the resulting distributions with Table 1. What can you observe?

Columnar Transposition

Besides substitution, it is also possible to encrypt a plaintext by permuting its characters. In ancient Greece, a device called a "scytale" was used. It consisted of a rod and a strip (e.g., of leather) with the message. The strip was wrapped around the rod. To decrypt the message, knowledge of the rod's diameter was required.

A more modern transposition method is called columnar transposition (Spaltenweise Transposition). There are several variants depending on read direction (column/row) and key convention. In this exercise, we illustrate one variant using an example.

Given the plaintext: Beispiele.
Encrypt with the key: HAL
The text is arranged into columns (3 columns because "HAL" has 3 letters):

B	E	I
S	P	I
E	L	E

Alphabetically sorting the key HAL gives A H L. This means column 2 (A) comes first, then column 1 (H), then column 3 (L). The resulting ciphertext is: EBIPSILEE

E	B	I
P	S	I
L	E	E

Exercise 2.

1. Based on the example above, think about how to decrypt this columnar transposition variant.

2. Decrypt the ciphertext: YRCOTPCSILOO using the key SEC. Describe your process.

Polyalphabetic Substitution - Vigenère

Instead of using a single alphabet like in Caesar, it's possible to use multiple alphabets. A popular method in the Middle Ages called Vigenère uses a keyword, e.g., "Sicher". Encryption and decryption work similarly to the Caesar cipher. The keyword **S**(K=18) **I**(K=8) **C**(K=2) **H**(K=7) **E**(K=4) **R**(K=17) determines the Caesar shift for each character in the plaintext. If the plaintext is longer than the key, the key repeats cyclically. Thus, character 7 uses shift 18 again, and so on.

For encryption/decryption, the Vigenère square is commonly used:

[1](a good source for German texts is `https://www.projekt-gutenberg.org/`)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Figure 1: Vigenère Square

Exercise 3.

1. Manually (without Cryptool), use the Vigenère square to encrypt: Cybersicherheitsvorlesung
2. Use Cryptool to encrypt various German texts[2] using the key: **SICHER**
3. Perform a frequency analysis on the ciphertexts using Cryptool. Compare your findings with those from Task 1. If you notice differences, describe potential reasons.

[2](a good source for German texts: `https://www.projekt-gutenberg.org/`)